

ANOVA HDMR: from DOE to PCE

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Outline

Design of Experiment

- Sobol' ANOVA HDMR

- Full Factorial Design

- Variance-based sensitivity indices

- Computing the coefficients

- Matrix of the model

PCE

- Legendre polynomials

- PCE vs FFD ANOVA representation

$$x_i \sim \mathcal{DU}(a_i, b_i)$$

The two-level full factorial design (R.A. Fisher)

Notations & Assumptions

- ▶ $y = \mathcal{M}(\mathbf{x})$ the scalar QoI
- ▶ $\mathbf{x} = (x_1, x_2, \dots, x_d)$
- ▶ $x_i \in [a_i, b_i] \sim \mathcal{DU}(a_i, b_i)$
- ▶ $X_i \in [-1, +1] \sim \mathcal{DU}(-1, +1)$
- ▶ $\mathbf{u} \sim \mathcal{U}(0, 1)^d$

Assumptions: $\mathbb{E}[y^2] = \int_{(0,1)^d} f^2(\mathbf{u}) d\mathbf{u} = \int_{\mathbb{R}^d} \mathcal{M}^2(\mathbf{x}) p_{\mathbf{x}} d\mathbf{x} < \infty$

$p_{\mathbf{x}} = \prod_{i=1}^d p_{x_i} \rightarrow$ independence

ANOVA HDMR

Under the previous assumptions, Sobol' (1993) shows that it is possible to cast $f(\mathbf{u})$ as follows,

$$f(\mathbf{u}) = f_0 + \sum_{i_1=1}^d f_{i_1}(u_{i_1}) + \sum_{i_2>i_1}^d f_{i_1,i_2}(u_{i_1}, u_{i_2}) \cdots + g_{1,\dots,d}(u_1, \dots, u_d) \quad (1)$$

$$= \sum_{\alpha \subseteq \{1,\dots,d\}} f_{\alpha}(\mathbf{u}_{\alpha}) \quad (2)$$

with $f_{\emptyset} = f_0 = \mathbb{E}[f(\mathbf{u})]$, and, $\int_0^1 f_{\alpha}(\mathbf{u}_{\alpha}) du_k = 0, \forall k \in \alpha$

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As a result:

- ▶ **Sobol's hdmr is unique**
- ▶ **The f_{α} 's are pairwise $\perp$: $\mathbb{E}[f_{\alpha}f_{\beta}] = V_{\alpha}\delta_{\alpha\beta}$**

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Consequently, the total variance of y can be decomposed as a sum of partial variances,

$V_y = \sum_{\alpha \subseteq \{1,\dots,d\}} V_{\alpha}$ with $V_{\alpha} = \mathbb{E}[f_{\alpha}^2(\mathbf{u}_{\alpha})]$ and $V_{\emptyset} = 0$.

Full Factorial Design

Consider $x_i \sim \mathcal{DU}(a_i, b_i)$ and $y = \mathcal{M}(\mathbf{x})$. The x -variables only take 2 possible values (two level). This defines a finite space made of $N = 2^d$ possibilities.

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$$y = a_0 + \sum_{i_1=1}^d a_{i_1} X_{i_1} + \sum_{i_2 > i_1}^d a_{i_1, i_2} X_{i_1} X_{i_2} + \dots + a_{1, \dots, d} X_1 X_2 \dots X_d \quad (3)$$

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We note that this is an ANOVA representation because:

$$\mathbb{E}[X_i] = \int X_i p_{X_i} dX_i = (-1) \times \frac{1}{2} + (+1) \times \frac{1}{2} = 0$$

As a consequence, the elements in (3) are **orthogonal** to each other.

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Besides, we note that:

$$\mathbb{E}[X_i^2] = \int X_i^2 p_{X_i} dX_i = (-1)^2 \times \frac{1}{2} + (+1)^2 \times \frac{1}{2} = 1$$

they are even **orthonormal** to each other.

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As a consequence:

$$V_y = \sum_{i_1=1}^d a_{i_1}^2 + \sum_{i_2 > i_1}^d a_{i_1, i_2}^2 + \dots + a_{1, \dots, d}^2 \quad (4)$$

Variance-based sensitivity indices

Sobol' ANOVA

$$S_i = \frac{V_i}{V_y}$$

FFD ANOVA

$$= \frac{a_i^2}{\sum_{\alpha \subseteq \{1, \dots, d\}} a_\alpha^2 - a_0^2}$$

Sensitivity indices

first-order effect

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first-order interaction

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$(k - 1)$ -th order interaction

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$(k - 1)$ -th order interaction

$$T_i = \frac{\sum_{\alpha_i \subseteq \mathcal{D}_{+i}} V_{\alpha_i}}{V_y}$$

$$= \frac{\sum_{\alpha_i \subseteq \mathcal{D}_{+i}} a_{\alpha_i}^2}{\sum_{\alpha \subseteq \{1, \dots, d\}} a_\alpha^2 - a_0^2}$$

total-order effect

$$\mathcal{D}_{+i} \subseteq \{1, \dots, d\} \text{ s.t. } \mathcal{D}_{+i} \cap \{i\} \neq \emptyset$$

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total-order effect

$$Sh_i = \sum_{\alpha_i \subseteq \mathcal{D}_{+i}} \frac{S_{\alpha_i}}{|\alpha_i|}$$

Shapley effect

Full Factorial Design

Let denote $\mathbf{X}$, a $N \times d$ matrix, the full factorial design associated with $X_i \sim \mathcal{DU}(-1, +1)$ and $\mathbf{y} = \{y_n\}_{n=1, \dots, N}$ the associated model responses.

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Let $\mathbf{M}$, a $N \times N$, be the matrix model associated with the FFD ANOVA, then

$$\mathbf{y} = \mathbf{M}\mathbf{a}$$

where $\mathbf{a}$ contains the **unknown** polynomial coefficients. Note $|\mathbf{a}| = N = 2^d$: Number of unknowns=Number of information.

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The polynomial coefficients $\mathbf{a}$ of the FFD ANOVA representation can be computed by OLS as follows:

$$\mathbf{a} = (\mathbf{M}^t \mathbf{M})^{-1} \mathbf{M}^t \mathbf{y} = \frac{\mathbf{M}^t \mathbf{y}}{N} \quad (5)$$

Because of the orthonormality of $(1, X_{i_1}, X_{i_1} X_{i_2}, \dots, X_1 \dots X_d)$, we have $\mathbf{M}^t \mathbf{M} = N \mathbf{I}_d$.

Full Factorial Design

Case $d = 3$: The full factorial design is ($N = 2^3 = 8$)

$$\mathbf{X} = \begin{bmatrix} -1 & -1 & -1 \\ -1 & -1 & +1 \\ -1 & +1 & -1 \\ -1 & +1 & +1 \\ +1 & -1 & -1 \\ +1 & -1 & +1 \\ +1 & +1 & -1 \\ +1 & +1 & +1 \end{bmatrix}$$

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The associated polynomial model is

$$y = a_0 + a_1X_1 + a_2X_2 + a_3X_3 + a_{1,2}X_1X_2 + a_{1,3}X_1X_3 + a_{2,3}X_2X_3 + a_{1,2,3}X_1X_2X_3$$

The model matrix associated with

$a_0 \times \mathbf{1} + a_1 X_1 + a_2 X_2 + a_3 X_3$, setting $\mathbf{a} = (a_0, a_1, a_2, a_3)$ is,

$$M = \begin{bmatrix} +1 & -1 & -1 & -1 \\ +1 & -1 & -1 & +1 \\ +1 & -1 & +1 & -1 \\ +1 & -1 & +1 & +1 \\ +1 & +1 & -1 & -1 \\ +1 & +1 & -1 & +1 \\ +1 & +1 & +1 & -1 \\ +1 & +1 & +1 & +1 \end{bmatrix}$$

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and so forth...

Setting $\mathbf{a} = (a_0, a_1, a_2, a_3, a_{1,2}, a_{1,3}, a_{2,3}, a_{1,2,3})$ the associated model matrix is then,

$$\mathbf{M} = \begin{bmatrix} +1 & -1 & -1 & -1 & +1 & +1 & +1 & -1 \\ +1 & -1 & -1 & +1 & +1 & -1 & -1 & +1 \\ +1 & -1 & +1 & -1 & -1 & +1 & -1 & +1 \\ +1 & -1 & +1 & +1 & -1 & -1 & +1 & -1 \\ +1 & +1 & -1 & -1 & -1 & -1 & +1 & +1 \\ +1 & +1 & -1 & +1 & -1 & +1 & -1 & -1 \\ +1 & +1 & +1 & -1 & +1 & -1 & -1 & -1 \\ +1 & +1 & +1 & +1 & +1 & +1 & +1 & +1 \end{bmatrix}$$

Exercise

The function models water flow through a borehole. The response of interest φ is the flow rate (in m^3/yr) defined as follows,

$$\varphi(\mathbf{x}) = \frac{2\pi T_u (H_u - H_l)}{\ln(r/r_w) \left(1 + \frac{2LT_u}{\ln(r/r_w)r_w^2 K_w} + \frac{T_u}{T_l} \right)}$$

The uncertain input variables, assumed **independent** of each other, are $\mathbf{x} = (r_w, r, T_u, H_u, T_l, H_l, L, K_w)$. Their marginal pdfs are given below,

r_w	Radius of Borehole (m)	[0.05, 0.15]
r	Radius of Influence (m)	[100, 50 000]
T_u	Transmissivity of Upper Aquifer (m^2/yr)	[63 070, 115 600]
H_u	Pressure Head of Upper Aquifer (m)	[990, 1 110]
T_l	Transmissivity of Lower Aquifer (m^2/yr)	[63.1, 116]
H_l	Pressure Head of Lower Aquifer (m)	[700, 820]
L	Length of Borehole (m)	[1 120, 1 680]
K_w	Hydraulic Conductivity (m/yr)	[9 855, 12 045]

Estimate the variance-based sensitivity indices with the FFD.

$$x_i \sim p_{x_i}$$

Polynomial Chaos Expansion (N. Wiener)

Notations & Assumptions

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- ▶ $\mathbf{x} = (x_1, x_2, \dots, x_d) \sim p_{\mathbf{x}}$
- ▶ $x_i \sim p_{x_i} = \frac{dF_{x_i}}{dx_i}$, the marginal pdf (F_{x_i} cdf) of x_i
- ▶ $u_i \sim \mathcal{U}(0, 1)$
- ▶ $\mathcal{D} = \{0, 1, \dots, d\}$, $\mathcal{D}_{-i} \cap \{i\} = \emptyset$, $\mathcal{D}_{+i} \cap \{i\} \neq \emptyset$

Assumptions: $\mathbb{E} [\mathcal{M}^2] = \int_{\mathbb{R}^d} \mathcal{M}^2(\mathbf{x}) p_{\mathbf{x}} d\mathbf{x} < \infty$ and $p_{\mathbf{x}} = \prod_{i=1}^d p_{x_i}$

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Legendre polynomials:

- ▶ $L_0 = 1$
- ▶ $L_1(u_i) = \sqrt{3}(2u_i - 1)$, first-degree polynomial
- ▶ $L_2(u_i) = \frac{\sqrt{5}}{2} (3(2u_i - 1)^2 - 1)$, second-degree polynomial ...

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Multivariate Legendre polynomials:

A multivariate polynomial basis is simply:

$\Psi_{\alpha}(\mathbf{u}) = L_{\alpha_1}(u_1)L_{\alpha_2}(u_2)\dots L_{\alpha_d}(u_d)$ the degree of which is $|\alpha| = \sum \alpha_k$ with $\alpha = \alpha_1\alpha_2\dots\alpha_d \in \mathbb{N}^d$.

PCE representation:

$$f(\mathbf{u}) = \sum_{\alpha \in \mathbb{N}^d} a_{\alpha} \Psi_{\alpha}(\mathbf{u}) \quad (6)$$

Reformulation: FFD ANOVA representation

We've just seen that if $y = \mathcal{M}(\mathbf{x})$ is square-integrable and that $x_i \sim \mathcal{DU}(a_i, b_i)$, by introducing $X_i \sim \mathcal{DU}(-1, 1)$, y can be cast as follows,

$$y = a_0 + \sum_{i_1=1}^d a_{i_1} X_{i_1} + \sum_{i_2 > i_1}^d a_{i_1, i_2} X_{i_1} X_{i_2} + \cdots + a_{1, \dots, d} X_1 X_2 \dots X_d$$

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Setting $P_0 = 1$, $P_1(X_i) = X_i$ and

$$\Phi_{\alpha}(\mathbf{X}) = \prod_{i=1}^d P_{\alpha_i}(X_i), \quad \alpha = \alpha_1 \alpha_2 \dots \alpha_d, \quad \alpha_i \in \{0, 1\}$$

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we can reformulate the FFD ANOVA representation as follows,

$$y = \sum_{\alpha \in \{0,1\}^d} a_{\alpha} \Phi_{\alpha}(\mathbf{X}) \quad (7)$$

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This is similar to Eq.(6), therefore it is a special case of the PCE.

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The PCE coefficients and variance-based sensitivity indices are computed similarly

Variance-based sensitivity indices

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$$S_i = \frac{V_i}{V_y}$$

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$$= \frac{a_i^2}{\sum_{\alpha \subseteq \{1, \dots, d\}} a_\alpha^2 - a_0^2}$$

Sensitivity indices

first-order effect

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$$= \frac{a_{i_1 i_2}^2}{\sum_{\alpha \subseteq \{1, \dots, d\}} a_\alpha^2 - a_0^2}$$

first-order interaction

$$S_{i_1 \dots i_k} = \frac{V_{i_1 \dots i_k}}{V_y}$$

$$= \frac{a_{i_1 \dots i_k}^2}{\sum_{\alpha \subseteq \{1, \dots, d\}} a_\alpha^2 - a_0^2}$$

$(k-1)$ -th order interaction

$$T_i = \sum_{\alpha_i \subseteq \mathcal{D}_{+i}} S_{\alpha_i}$$

$$= \frac{\sum_{\alpha_i \ni i} a_{\alpha_i}^2}{\sum_{\alpha \subseteq \{1, \dots, d\}} a_\alpha^2 - a_0^2}$$

total-order effect

$$Sh_i = \sum_{\alpha_i \subseteq \mathcal{D}_{+i}} \frac{S_{\alpha_i}}{|\alpha_i|}$$

Shapley effect

Uncertainty analysis of the borehole model

taken from <https://www.sfu.ca/~ssurjano/>

Practicum

The function models water flow through a borehole. The response of interest φ is the flow rate (in m^3/yr) defined as follows,

$$\varphi(\mathbf{x}) = \frac{2\pi T_u (H_u - H_l)}{\ln(r/r_w) \left(1 + \frac{2LT_u}{\ln(r/r_w)r_w^2 K_w} + \frac{T_u}{T_l} \right)}$$

The uncertain input variables, assumed **independent** of each other, are $\mathbf{x} = (r_w, r, T_u, H_u, T_l, H_l, L, K_w)$. Their marginal pdfs are given below,

r_w	Radius of Borehole (m)	$\mathcal{N}(r_w 0.11, 0.005^2)$
r	Radius of Influence (m)	$\mathcal{N}(\ln(r) 7.71, 1)$
T_u	Transmittivity of Upper Aquifer (m^2/yr)	$\mathcal{T}(T_u 63\,070, 115\,600, 100\,000)$
H_u	Pressure Head of Upper Aquifer (m)	$\mathcal{N}(\ln(H_u) 6.95, 0.0167^2)$
T_l	Transmittivity of Lower Aquifer (m^2/yr)	$\mathcal{U}(T_l 63, 116)$
H_l	Pressure Head of Lower Aquifer (m)	$\mathcal{N}(\ln(H_l) 6.6, 0.033^2)$
L	Length of Borehole (m)	$\mathcal{U}(L 1\,120, 1\,680)$
K_w	Hydraulic Conductivity (m/yr)	$\mathcal{U}(K_w 9\,855, 12\,045)$

Estimate the variance-based sensitivity indices of each variable.